

Research Interests

Sensorimotor Neuroscience · Robotics · Systems Identification and Synthesis · Generalizable Control · Perception

Education

Okinawa Institute of Science and Technology

Ph.D. in Computational Neuroscience

Okinawa, Japan

Jan. '25 — May '26

- Withdrew from the program in good standing to better pursue personal research interests and career goals
- Advisors: Greg Stephens, Amedeo Roberto Esposito
- Selected coursework: Neuromotor Control, Nonlinear Dynamical Systems, Brain Computation, Cognitive Neurorobotics

University of Western Ontario

M.Sc. in Neuroscience; Grade: A

London, Canada

Sept. '22 — Oct. '24

- Thesis title: The Nature of Reflexes in Online Planning and Control [\[link\]](#)
- Advisors: Andrew Pruszynski, Paul Gribble
- Recipient of the Vector Scholarship in Artificial Intelligence (C\$17.5k) and the BrainsCAN Graduate Studentship (C\$50k)
- Selected coursework: Principles of Neuroscience, Reinforcement Learning

University of Toronto

B.A.Sc. in Electrical Engineering; cGPA: 3.92/4.0

Toronto, Canada

Sept. '17 — Apr. '22

- Engineering International Scholar: received a full tuition fee waiver for the entire duration of the program (C\$229k)
- Recipient of the Adel S. Sedra Gold Medal for achieving the highest cumulative average in the graduating class
- Graduated with High Honors and Minor in Robotics and Mechatronics; Dean's Honor List in all semesters
- Selected coursework (*graduate-level): Linear Control Systems, *Random Processes, *Sensory Communication, Analog and Digital Electronics, System Mapping, Machine Learning, Real-time Control, Robot Modeling and Control, Mechatronics

Selected Experiences

Doctoral Researcher, OIST Graduate University

Biological Physics Theory Unit and Information Theory, Probability, and Statistics Unit

Okinawa, Japan

Jan. '25 — May '26

- Focusing on understanding closed-loop neural dynamics through behaviorally informative decomposition of control
- Completed several mini projects on topics like adaptive output chunking in recurrent neural networks, passive-active optimal control, dynamical systems reconstruction, speed-curvature power-law, and I/O stability of plastic spiking neural networks

Graduate Researcher, Western Institute for Neuroscience

Sensorimotor Superlab

London, Canada

Sept. '22 — Oct. '24

- Investigated goal-dependent coupling between reflexive and voluntary motor control by estimating update times of both modalities independently during ongoing movement; also tried investigating role of motor variability in implicit adaptation
- Led multiple discussions on motor planning and control; also mentored and trained undergraduate research assistants

Navigation Engineering Intern, Zebra Technologies

Path-Planning and Control Team

Mississauga, Canada

May '20 — Aug. '21

- Critiqued, implemented, and tested auto-navigation algorithms from literature for deployment on inventory scanning robots
- Redesigned path planner to improve aisle scan coverage and efficiency by robustly handling obstructions and curved aisles
- Developed tools for rapidly prototyping planners and controllers and benchmarking their performance; written in Julia
- Reviewed and fixed real-world behavior and performance bugs; identified, proposed, and applied planner improvements

Software Engineering Intern, Rocscience Inc.

Geotechnical Software Tools Design

Toronto, Canada

May '19 — Aug. '19

- Developed interactive geometry import and diagnostic tools for file formats external to, and also within, the software suite

Undergraduate Research Assistant, Motion Adaptation Science Lab

Toronto Rehabilitation Institute

Toronto, Canada

May '18 — Aug. '18

- Created finite element models of the lower leg from MRI data for optimizing functional electro-stimulation strategies

Talks & Posters

3. Integration Timescales of Visuospatial Task Constraints and Composition Strategies in Reflex Responses. Poster, Champalimaud Research Symposium: *Neurocybernetics at Scale*, Portugal, 2025. [\[pdf\]](#) Awarded a travel grant (€1500) by Champalimaud Foundation and Protocol Labs.

2. Stretch Reflexes Quickly Integrate Spatial Task Constraints During Reaching. Invited talk, Department of Cognitive and Brain Sciences, IIT Gandhinagar, and Institute of Neuroscience, UC Louvain, 2024.
1. Computing Error-bound Inverse Kinematics Solutions in Fixed-time using Low-Power Analog Circuits. Podium presentation, Undergraduate Engineering Research Day, University of Toronto, 2021. [🔗 \[slides\]](#)

Technical Skills

Python · Julia · Matlab · C/C++ · ROS · PyTorch · OpenCV · Typst · Graphics Design · Mechatronics · Machining

Selected Projects

Intuitive Robot Arm Reach: Undergraduate final-year project with the aim to develop a reach control method by combining algorithmic models of human motor-learning processes, such as proximodistal exploration and motor babbling. Designed and evaluated an interpolation-based method that covered the entire reachable space, contrary to tuning for specific reach targets.

Analog IK Solver: Presents a generic computation method for inverse kinematics of serial robots using a modular and digitally reconfigurable analog circuit. Error-bound approximate solutions are found in fixed time by continuously evaluating forward kinematics using periodic, phase-locked, and frequency-scaled waveforms as successive joint configurations.

CollabBots: A robotic platform, built from scratch, with two opposing robotic arms and orthogonal cameras to implement and test visual servoing and collaborative scheduling for picking and placing toy blocks in pre-specified goal patterns.

Miscellaneous

Academic Officer: Elected to represent the student assembly at OIST and liaise with the graduate school. Responsibilities included identifying inefficiencies and suggesting improvements in the academic program and advocating for student concerns and interests.

Big Ideas Committee: Led a club that is a part of the Society for Neuroscience Graduate Students (SONGS) at Western University. We hosted didactic discussions as well as regular share-what-you-read sessions on topics like creativity, philosophy of science, and consciousness. We also organized a student-led panel discussion on the scientific and philosophical study of consciousness, which was attended by 40+ graduate students and postdocs.

Hobbies and Side Interests: Photography and painting; citizen science; education systems; cognitive architectures; philosophy of mind, memory, and action; Indian philosophy; Sanskrit language; yoga and meditation.

Sports and Outdoor Activities: Love playing cricket and badminton, and have also played in local cricket leagues. I also enjoy running, hiking, camping, and long walks in nature.